

COMMUNICATION SYSTEMS 2 (COM261S): LAB REPORTS

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1. How to plot a graph on MATLAB

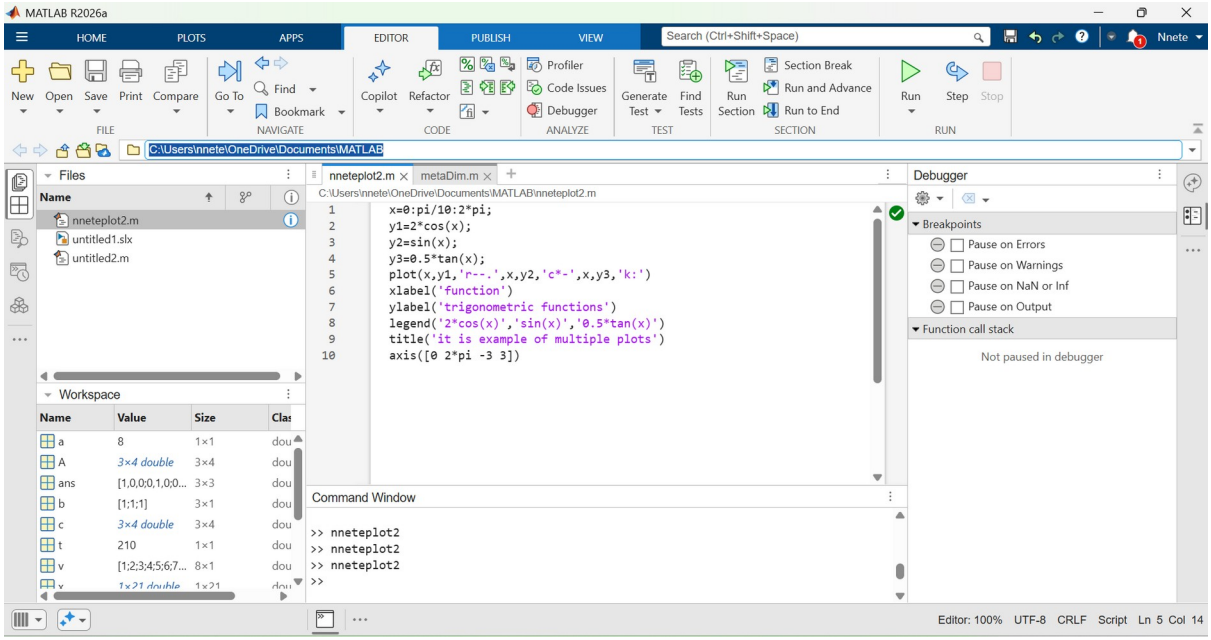


FIGURE 1: HOW TO PLOT A GRAPH SIMULATION INPUT

How to plot a graph Simulation Output 1:

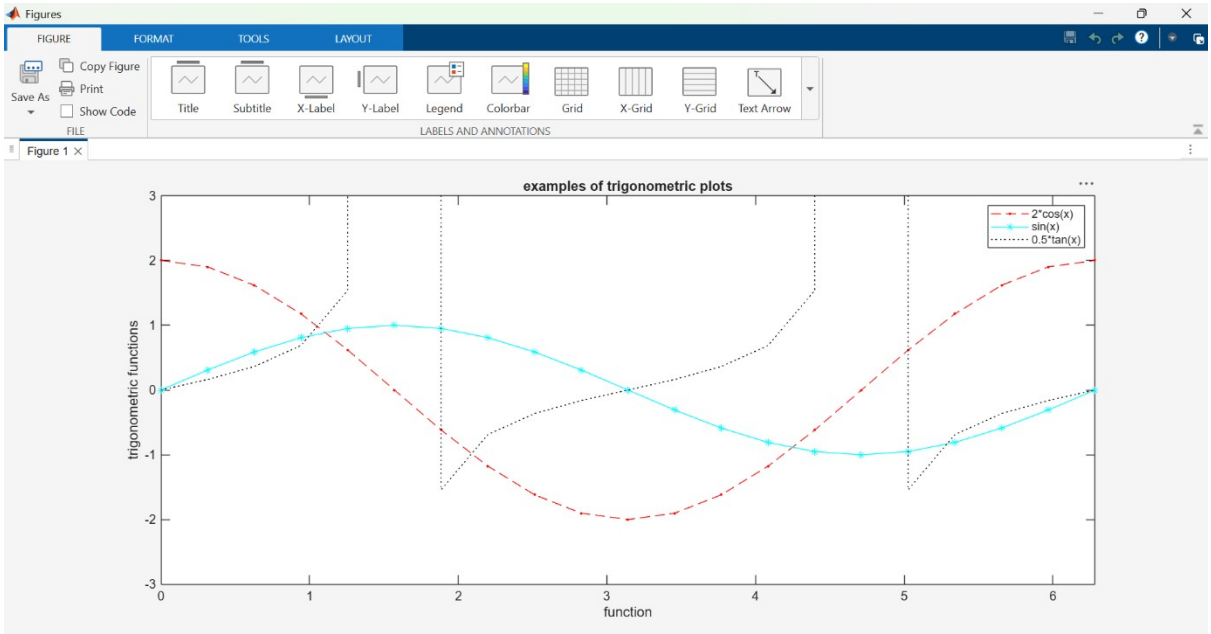


FIGURE 2: HOW TO PLOT A GRAPH SIMULATION OUTPUT 1

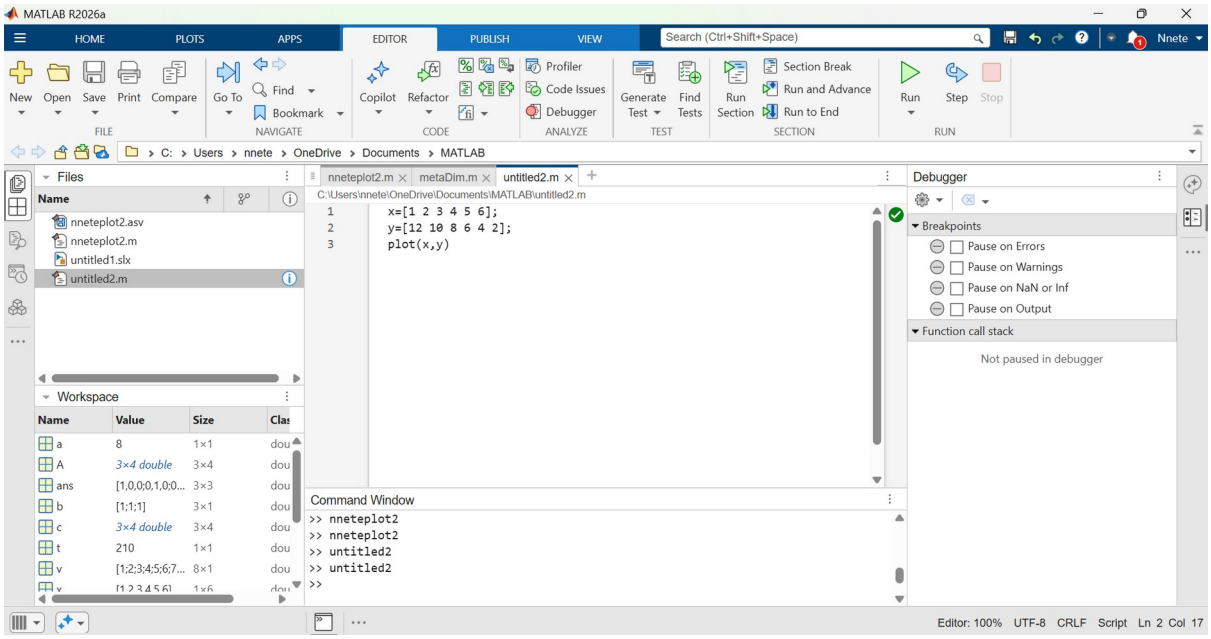


FIGURE 3: HOW TO PLOT A GRAPH SIMULATION INPUT 2

How to plot a graph Simulation output 2:

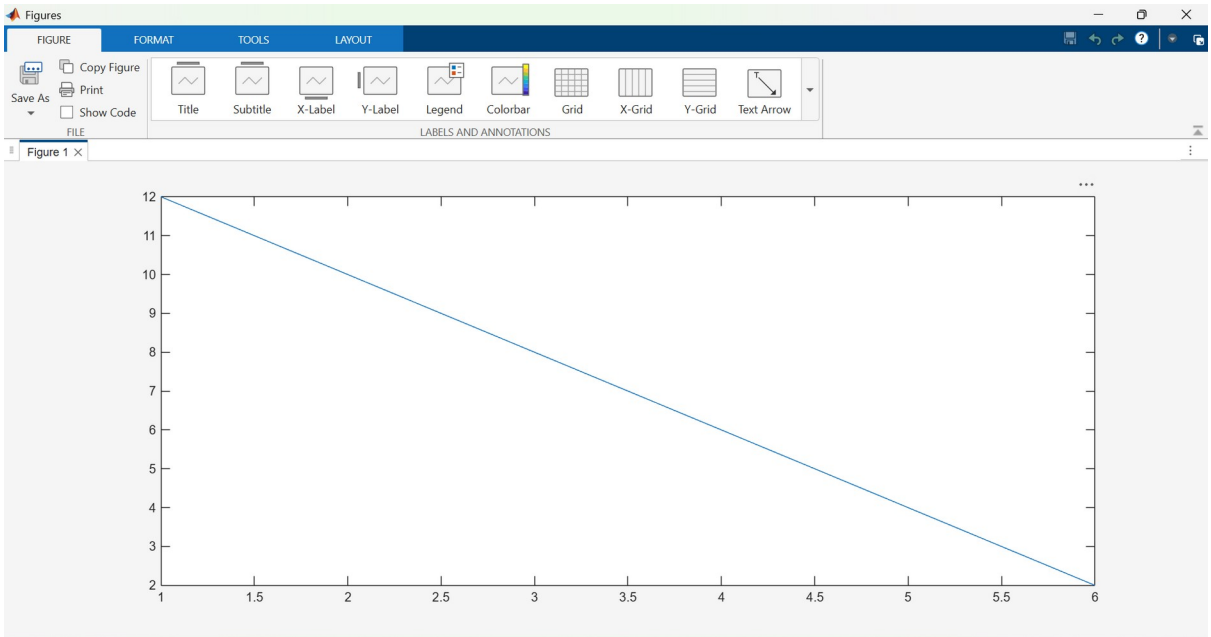


FIGURE 4: HOW TO PLOT A GRAPH SIMULATION OUTPUT 2

2. MATLAB As A Calculator

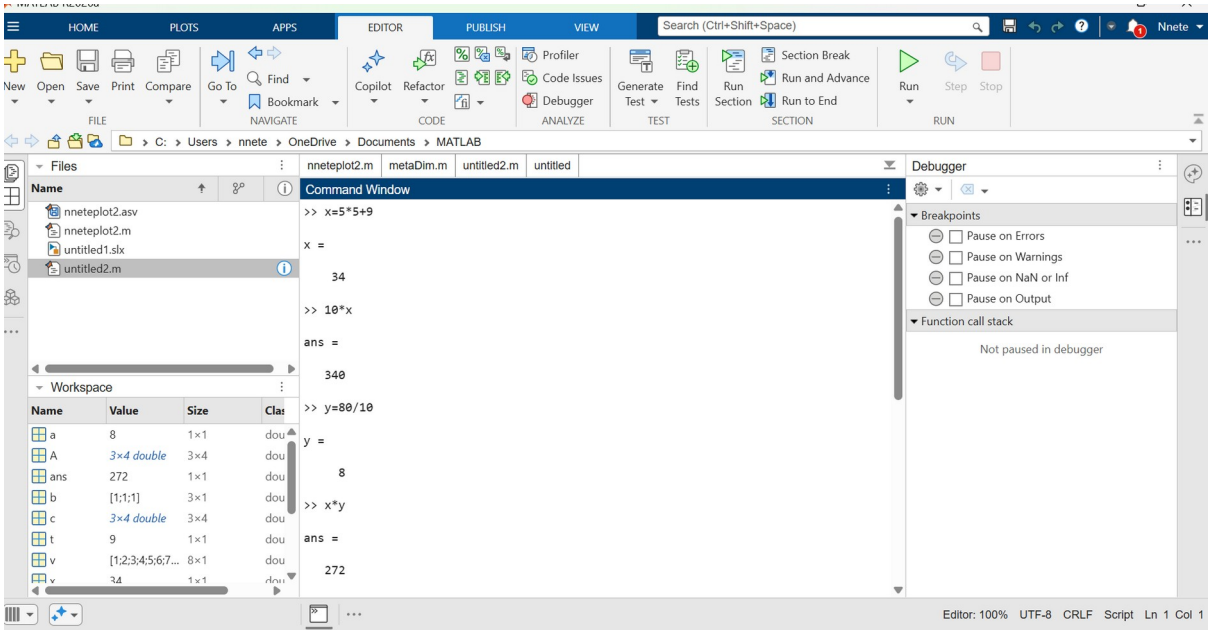


FIGURE 5: MATLAB AS A CALCULATOR SIMULATION 1

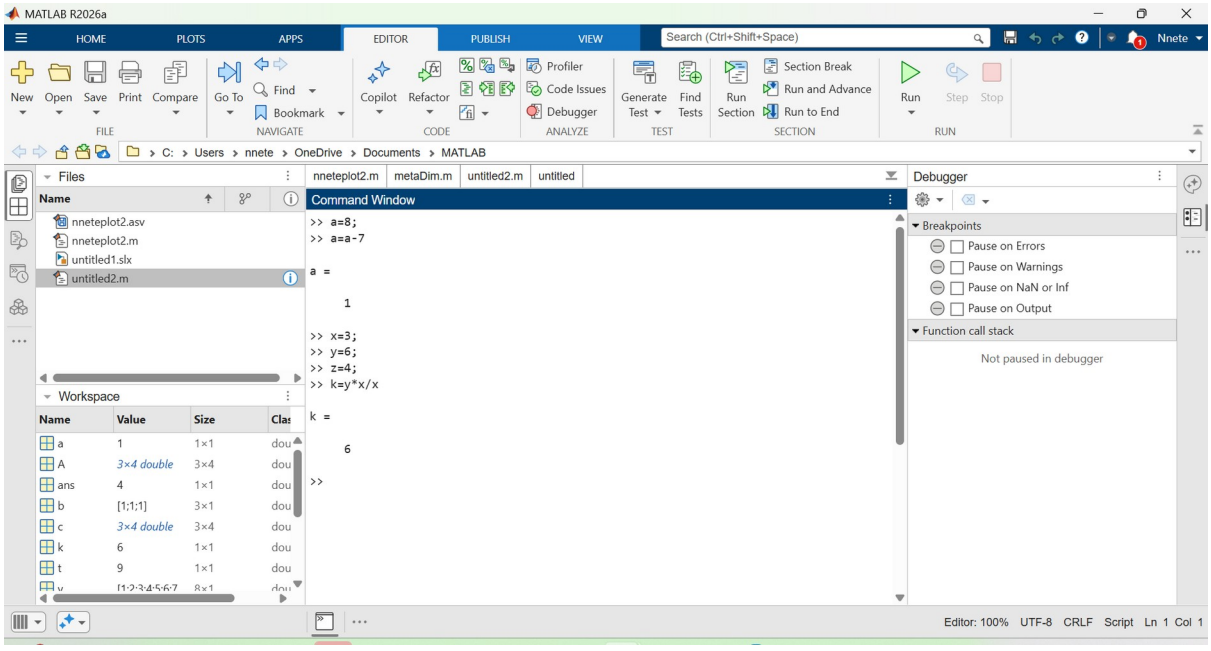


FIGURE 6:MATLAB AS A CALCULATOR SIMULATION 3



FIGURE 7: MATLAB AS A CALCULATOR SIMULATION 4

3. Matrix On MATLAB

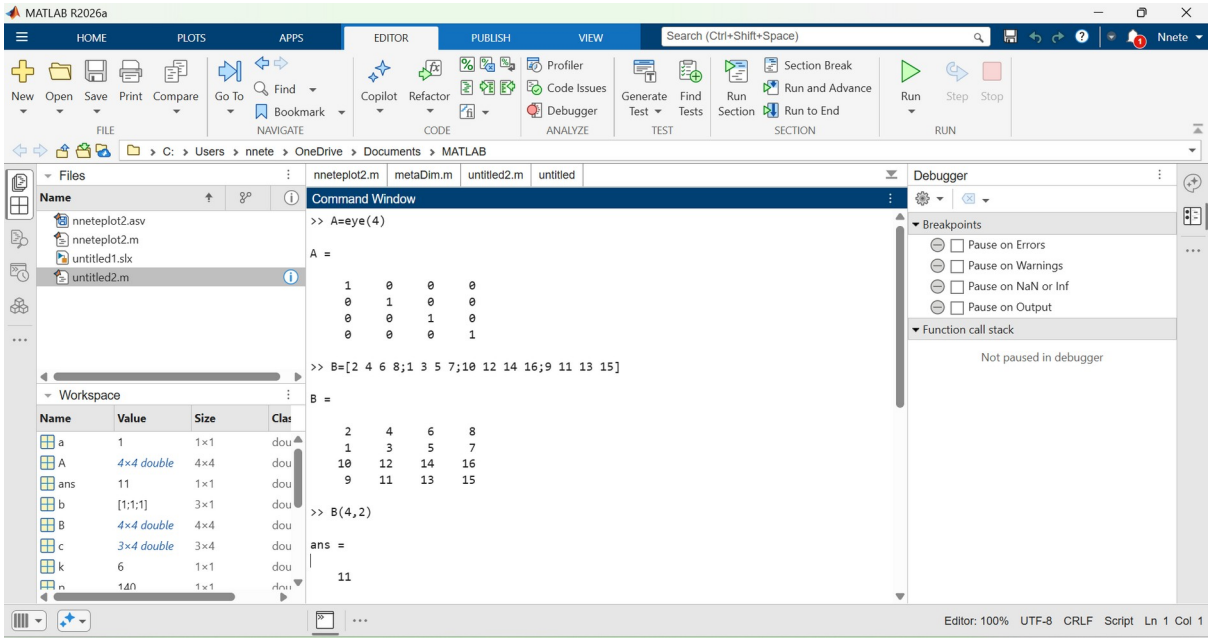


FIGURE 8:MATRIX ON MATLAB SIMULATION 1

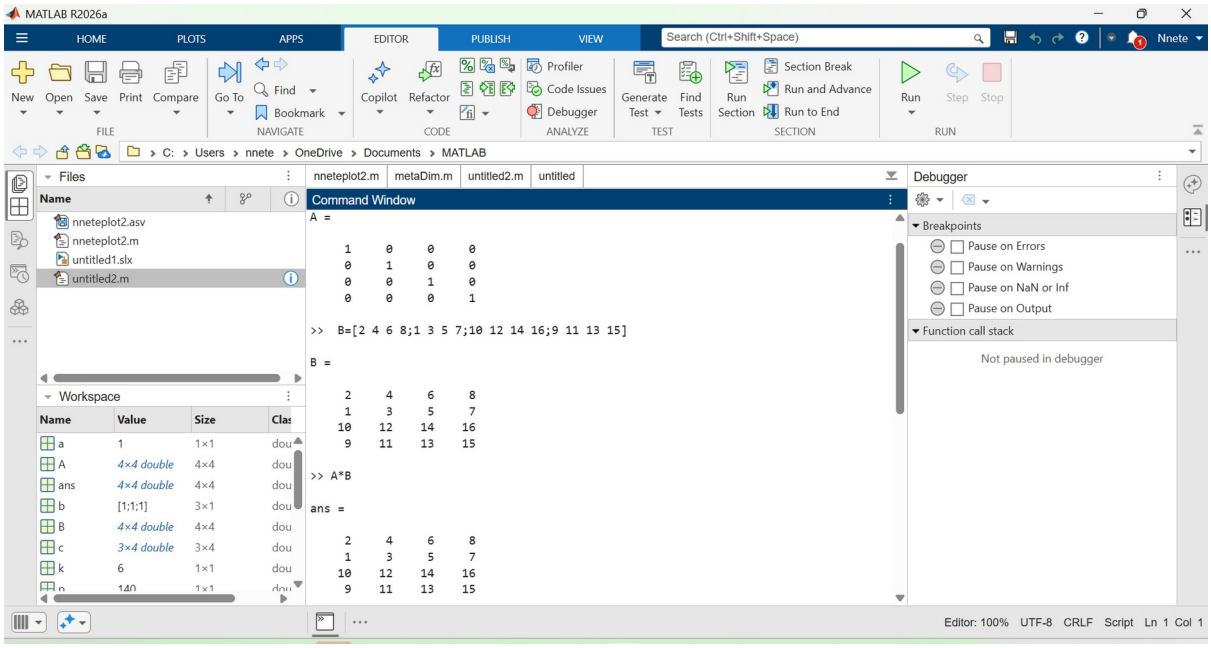


FIGURE 9: MATRIX ON MATLAB SIMULATION 2

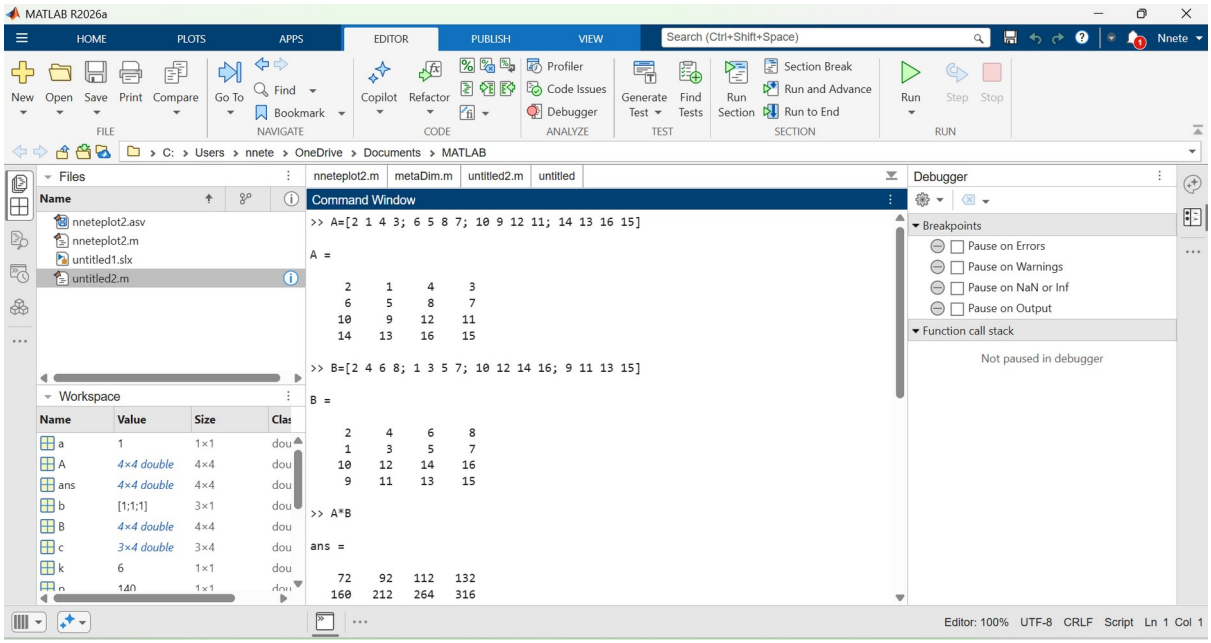


FIGURE 10: MATRIX ON MATLAB SIMULATION 3

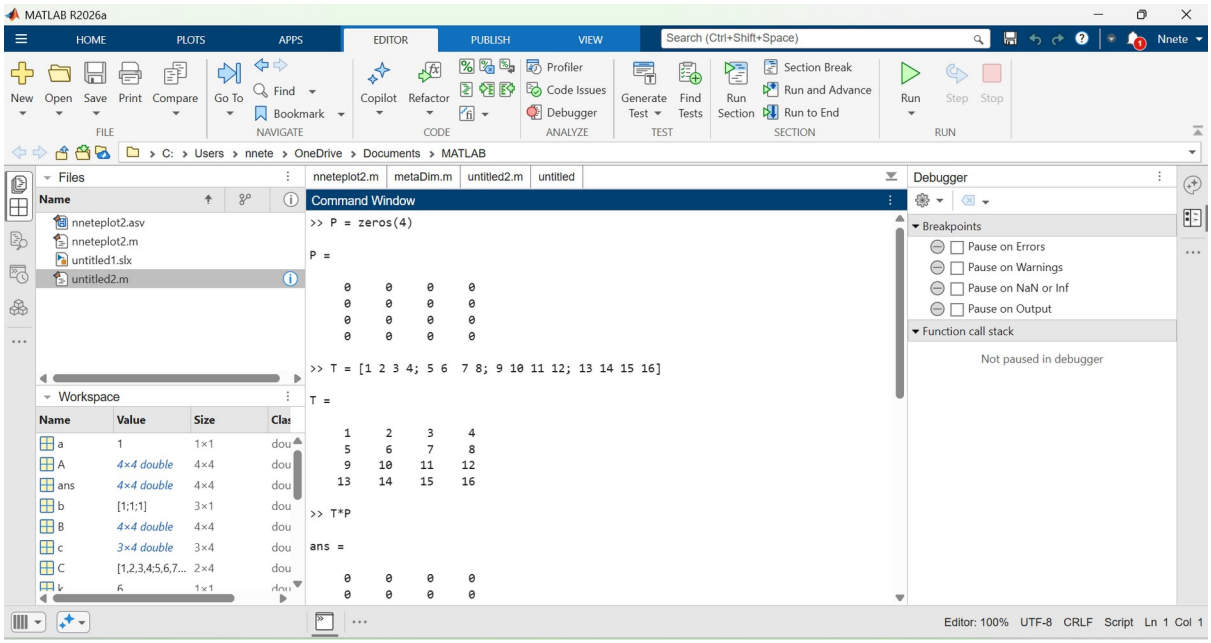


FIGURE 11: MATRIX ON MATLAB SIMULATION 4

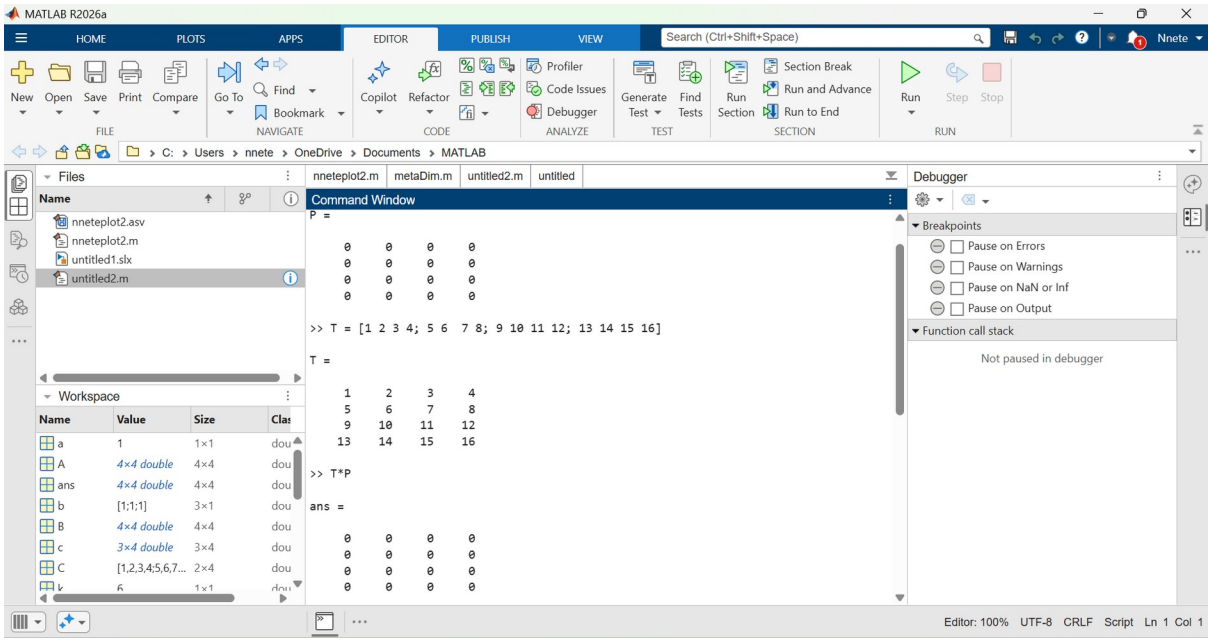


FIGURE 12: MATRIX ON MATLAB SIMULATION 5

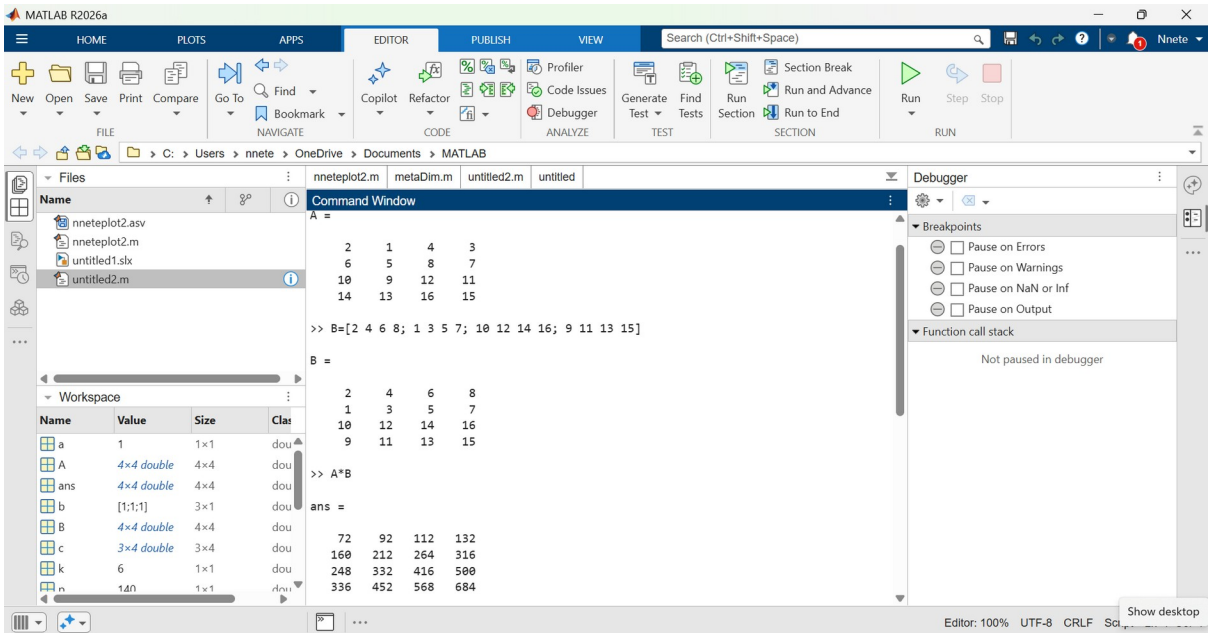


FIGURE 13: MATRIX ON MATLAB SIMULATION 6

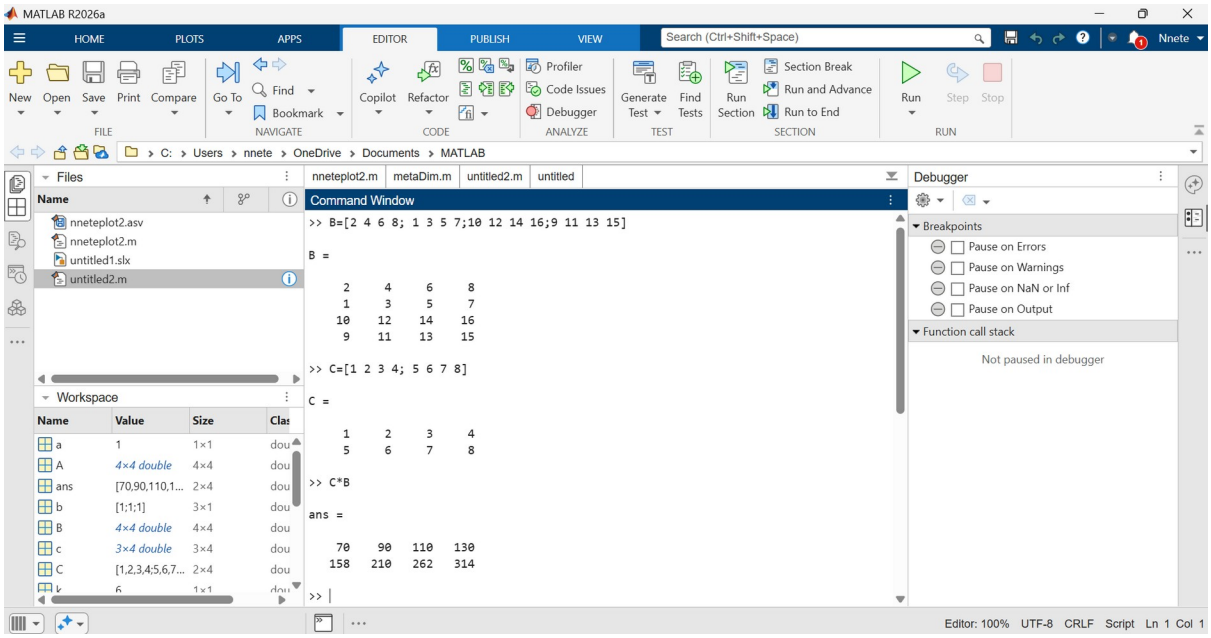


FIGURE 14: MATRIX ON MATLAB SIMULATION 7

4. Practical Simulations

4.1 Practical 1: Amplitude Modulation

```
clc;
clear all;
close all;

Ac=3; %carrier amplitude
fc=0.6; %carrier frequency
Am=.6; %message signal amplitude
fm=.06; %message signal frequency
Fs=110; %sampling rate/frequency
ka=1;%Amplitude Sensitivity

t=[0:0.1:50];%defining the time range & disseminating it into samples

ct=Ac*cos(2*pi*fc*t); %defining the carrier signal wave
mt=Am*cos(2*pi*fm*t); %defining the message signal
AM=ct.*(1+ka*mt); %Amplitude Modulated wave, according to the standard definition

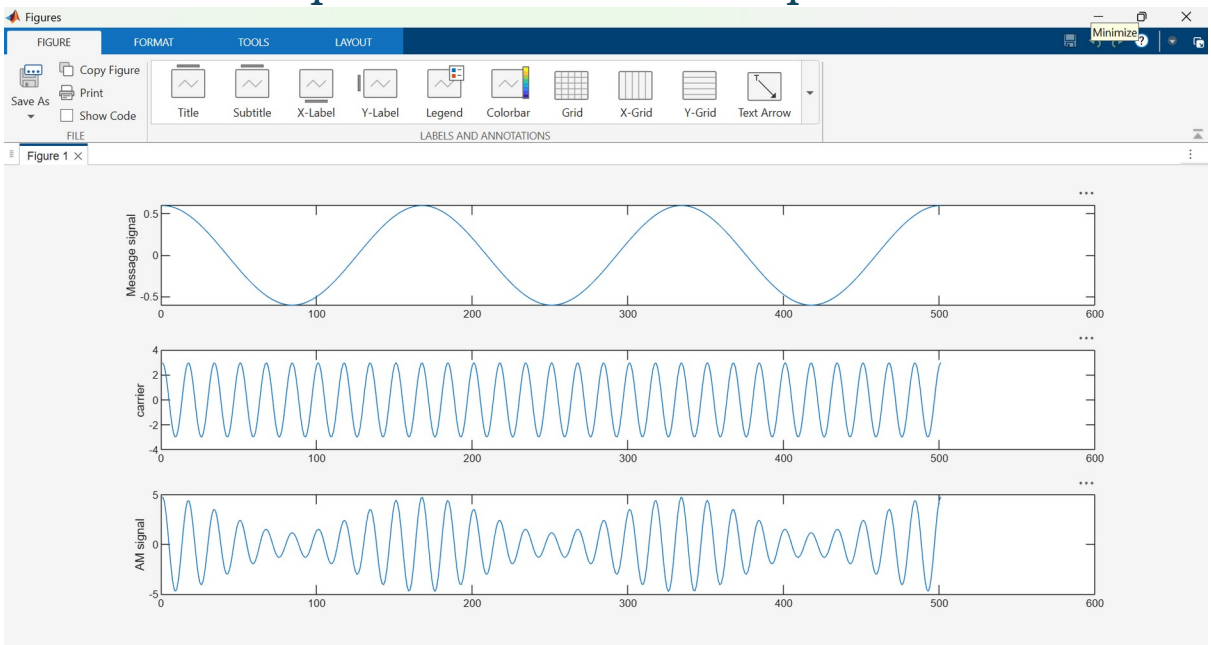
subplot(3,1,1);%plotting the message signal wave
plot(mt);
ylabel('Message signal');

subplot(3,1,2); %plotting the carrier signal wave
plot(ct);
ylabel('carrier');

subplot(3,1,3); %plotting the amplitude modulated wave
plot(AM);
ylabel('AM signal')
```

FIGURE 15: AMPLITUDE MODULATION CODE

Practical 1: Amplitude Modulation Output



4.2 Practical 2: Frequency Modulation

```
clc
clear all
close all

t = 0:0.001:1; %up to 1000 samples
vm = input('Enter Amplitude (Message) = ');
vc = input('Enter Amplitude (Carrier) = ');
fM = input('Enter Message frequency = ');
fc = input('Enter Carrier frequency = ');
m = input('Enter Modulation Index = ');

msg = vm*sin(2*pi*fM*t);

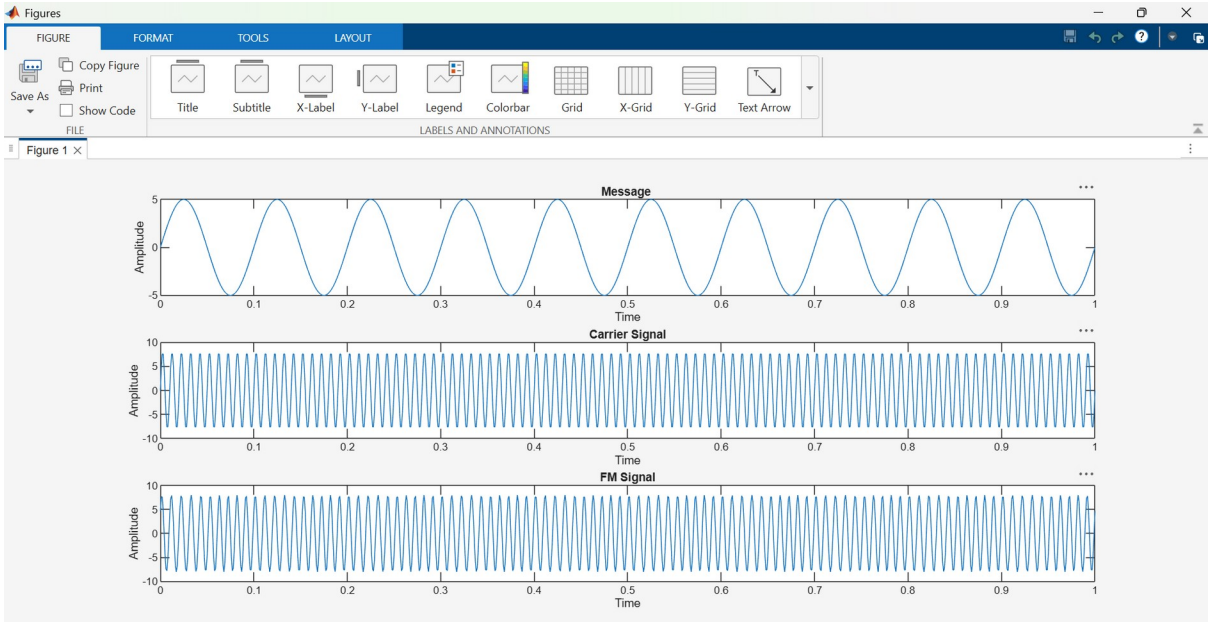
subplot(3,1,1); %plotting message signal
plot(t,msg);
xlabel('Time');
ylabel('Amplitude');
title('Message ');
carrier = vc*sin(2*pi*fc*t);

subplot(3,1,2); %plotting carrier signal
plot(t,carrier);
xlabel('Time');
ylabel('Amplitude');
title('Carrier Signal');
y = vc*sin(2*pi*fc*t+m.*cos(2*pi*fM*t));

subplot(3,1,3);%plotting FM (Frequency Modulated) signal
plot(t,y);
xlabel('Time');
ylabel('Amplitude');
title('FM Signal');
```

FIGURE 16: FREQUENCY MODULATION CODE

Practical 2: Frequency Modulation Output



4.3 Practical 3: ASK Modulation and Demodulation

```
Binary information at Reciver :
    1     0     1     0     1     1     1
```

FIGURE 15: ASK BINARY INFORMATION

```
clc;
clear all;
close all;
x=[ 1 0 1 0 1 1 1]; %Binary information
bp=.000001; %BIT period
disp(' Binary information at Transmitter :');
disp(x);
%Representation of transmitting binary information as digital signal
bit=[];
for n=1:1:length(x)
    if x(n)==1
        se=ones(1,100);
    else x(n)=0;
        se=zeros(1,100);
    end
    bit=[bit se];
end
t1=bp/100:bp/100:100*length(x)*(bp/100);
subplot(3,1,1);
plot(t1,bit,'lineWidth',2.5);grid on;
axis([ 0 bp*length(x) -.5 1.5]);
ylabel('amplitude(volt)');
xlabel(' time(sec)');
title('transmitting information as digital signal');


%Binary-ASK modulation
A1=10; % Amplitude of carrier signal for information 1
A2=5; % Amplitude of carrier signal for information 0
br=1/bp; % bit rate
f=br*10; % carrier frequency
t2=bp/99:bp/99:bp;
ss=length(t2);
m=[];
for (i=1:1:length(x))
    if (x(i)==1)
        y=A1*cos(2*pi*f*t2);
    else
        y=A2*cos(2*pi*f*t2);
    end
    m=[m y];
end
t3=bp/99:bp/99:bp*length(x);
subplot(3,1,2);
plot(t3,m);
xlabel('time(sec)');
ylabel('amplitude(volt)');
title('waveform for binary ASK modulation coresponding binary information');


%Binary ASK demodulation
mn=[];
for n=ss:ss:length(m)
    t=bp/99:bp/99:bp;
    y=cos(2*pi*f*t); % carrier siignal
    mm=y.*m((n-(ss-1)):n);
    t4=bp/99:bp/99:bp;
    z=trapz(t4,mm); % intregation
    zz=round((2*z/bp));
    if(zz>7.5) % logic level = (A1+A2)/2=7.5
        a=1;
```

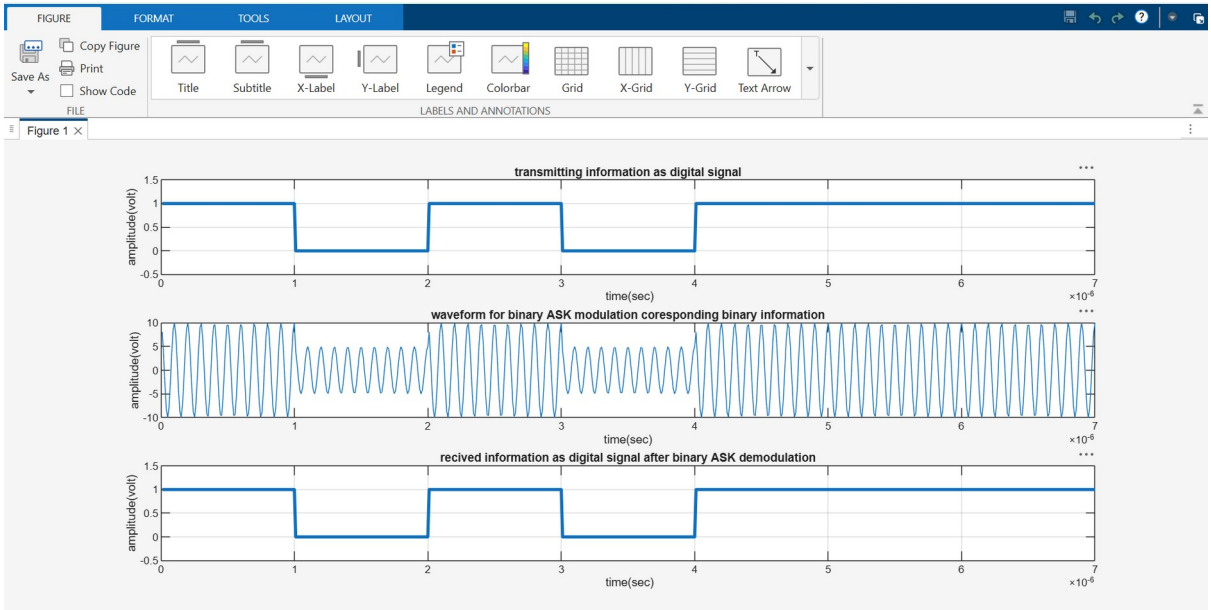
FIGURE 17: ASK MODULATION AND DEMODULATION CODE PART 1

```
    else
        a=0;
    end
    mn=[mn a];
end
disp(' Binary information at Reciver :');
disp(mn);


%Representation of binary information as digital signal which archived
%after ASK demodulation
bit=[];
for n=1:length(mn)
    if mn(n)==1
        se=ones(1,100);
    else mn(n)=0;
        se=zeros(1,100);
    end
    bit=[bit se];
end
t4=bp/100:bp/100:100*length(mn)*(bp/100);
subplot(3,1,3)
plot(t4,bit,'LineWidth',2.5);grid on;
axis([ 0 bp*length(mn) -.5 1.5]);
ylabel('amplitude(volt)');
xlabel(' time(sec)');
title('recived information as digital signal after binary ASK demodulation');
%end of program
```

FIGURE 18: ASK MODULATION AND DEMODULATION CODE PART 2

Practical 3: ASK Modulation and Demodulation Output



4.4 Practical 4: PSK Modulation and Demodulation

```
Binary information at Receiver :
    1     0     1     0     1     1     0
```

FIGURE 19: PSK MODULATION BINARY INFORMATION

```
clc;
clear all;
close all;
x=[ 1 0 1 0 1 1 0]; % Binary Information
bp=.000001; % bit period
disp(' Binary information at Transmitter :');
disp(x);

%Representation of transmitting binary information as digital signal
bit=[];
for n=1:1:length(x)
    if x(n)==1;
        se=ones(1,100);
    else x(n)==0;
        se=zeros(1,100);
    end
    bit=[bit se];
end
t1=bp/100:bp/100:100*length(x)*(bp/100);
subplot(3,1,1);
plot(t1,bit,'lineWidth',2.5);grid on;
axis([ 0 bp*length(x) -.5 1.5]);
ylabel('amplitude(volt)');
xlabel(' time(sec)');
title('transmitting information as digital signal');

%Binary-PSK modulation
A=5; % Amplitude of carrier signal
br=1/bp; % bit rate
f=br*2; % carrier frequency
t2=bp/99:bp/99:bp;
ss=length(t2);
m=[];
for (i=1:1:length(x))
    if (x(i)==1)
        y=A*cos(2*pi*f*t2);
    else
        y=A*cos(2*pi*f*t2+pi); %A*cos(2*pi*f*t+pi) means -A*cos(2*pi*f*t)
    end
    m=[m y];
end
t3=bp/99:bp/99:bp*length(x);
subplot(3,1,2);
plot(t3,m);
xlabel('time(sec)');
ylabel('amplitude(volt)');
title('waveform for binary PSK modulation corresponding the binary information');

%Binary PSK demodulation
mn=[];
for n=ss:ss:length(m)
    t=bp/99:bp/99:bp;
    y=cos(2*pi*f*t); % carrier signal
    mm=y.*m((n-(ss-1)):n);
    t4=bp/99:bp/99:bp;
    z=trapz(t4,mm) % integration
    zz=round((2*z/bp))
    if(zz>0) % logic level = (A+A)/2=0
        %because A*cos(2*pi*f*t+pi) means -A*cos(2*pi*f*t)
        a=1;
    else
```

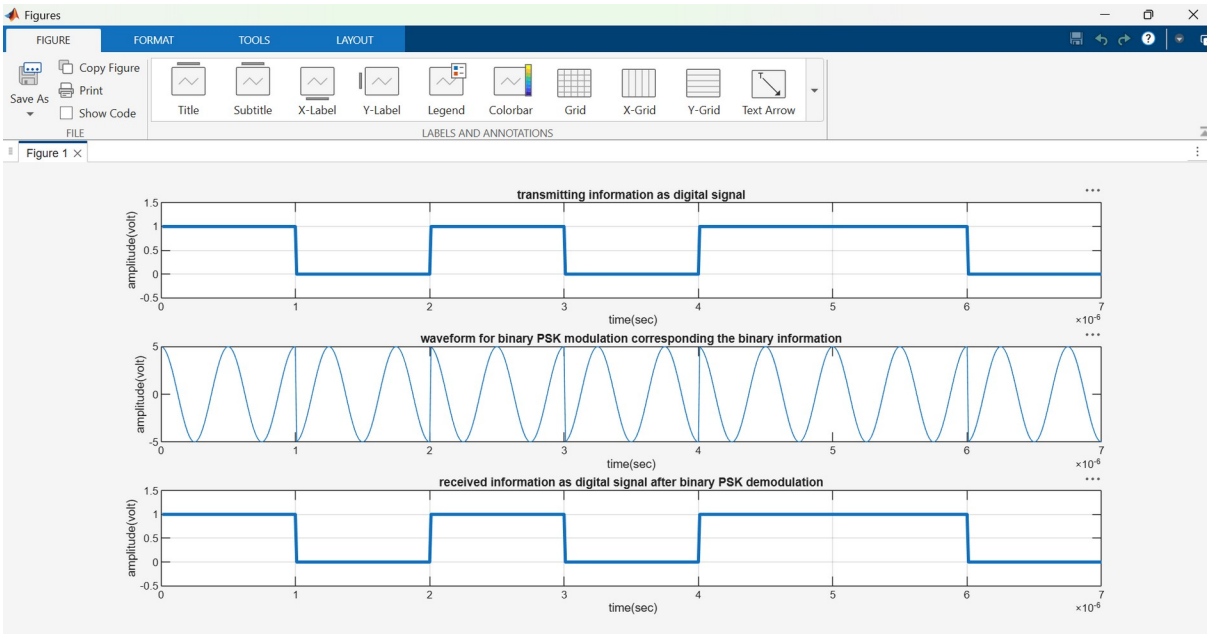
FIGURE 20: PSK MODULATION AND DEMODULATION CODE PART 1

```
        a=0;
    end
    mn=[mn a];
end
disp(' Binary information at Receiver :');
disp(mn);
%Representation of binary information as digital signal which archived

%After PSK demodulation
bit=[];
for n=1:length(mn);
    if mn(n)==1;
        se=ones(1,100);
    else mn(n)==0;
        se=zeros(1,100);
    end
    bit=[bit se];
end
t4=bp/100:bp/100:100*length(mn)*(bp/100);
subplot(3,1,3)
plot(t4,bit,'LineWidth',2.5);grid on;
axis([ 0 bp*length(mn) -.5 1.5]);
ylabel('amplitude(volt)');
xlabel(' time(sec)');
title('received information as digital signal after binary PSK demodulation');
```

FIGURE 21: PSK MODULATION AND DEMODULATION CODE PART 2

Practical 4: PSK Modulation and Demodulation Output



4.5 Practical 5: FSK Modulation and Demodulation

```
Binary information at Receiver :
    1     0     1     0     1     1     0
```

FIGURE 22: FSK MODULATION BINARY INFORMATION

```
clc;
clear all;
close all;
x=[ 1 0 1 0 1 1 0]; % Binary Information
bp=.000001; % bit period
disp(' Binary information at Trans mitter :');
disp(x);

%Representation of transmitting binary information as digital signal
bit=[];
for n=1:1:length(x)
    if x(n)==1
        se=ones(1,100);
    else x(n)=0;
        se=zeros(1,100);
    end
    bit=[bit se];
end
t1=bp/100:bp/100:100*length(x)*(bp/100);
subplot(3,1,1);
plot(t1,bit,'lineWidth',2.5);grid on;
axis([ 0 bp*length(x) -.5 1.5]);
ylabel('amplitude(volt)');
xlabel(' time(sec)');
title('transmitting information as digital signal');

%Binary FSK modulation
A=5; % Amplitude of carrier signal
br=1/bp; % bit rate
f1=br*8; % carrier frequency for information as 1
f2=br*2; % carrier frequency for information as 0
t2=bp/99:bp/99:bp;
ss=length(t2);
m=[];
for (i=1:1:length(x))
    if (x(i)==1)
        y=A*cos(2*pi*f1*t2);
    else
        y=A*cos(2*pi*f2*t2);
    end
    m=[m y];
end
t3=bp/99:bp/99:bp*length(x);
subplot(3,1,2);
plot(t3,m);
xlabel('time(sec)');
ylabel('amplitude(volt)');
title('waveform for binary FSK modulation corresponding binary information');

%Binary FSK demodulation
mn=[];
for n=ss:ss:length(m)
    t=bp/99:bp/99:bp;
    y1=cos(2*pi*f1*t); % carrier signal for information 1
    y2=cos(2*pi*f2*t); % carrier signal for information 0
    mm=y1.*m((n-(ss-1)):n);
    mmm=y2.*m((n-(ss-1)):n);
    t4=bp/99:bp/99:bp;
    z1=trapz(t4,mm) % integration
    z2=trapz(t4,mmm) % integration
    zz1=round(2*z1/bp)
```

FIGURE 23: FSK MODULATION AND DEMODULATION CODE PART 1

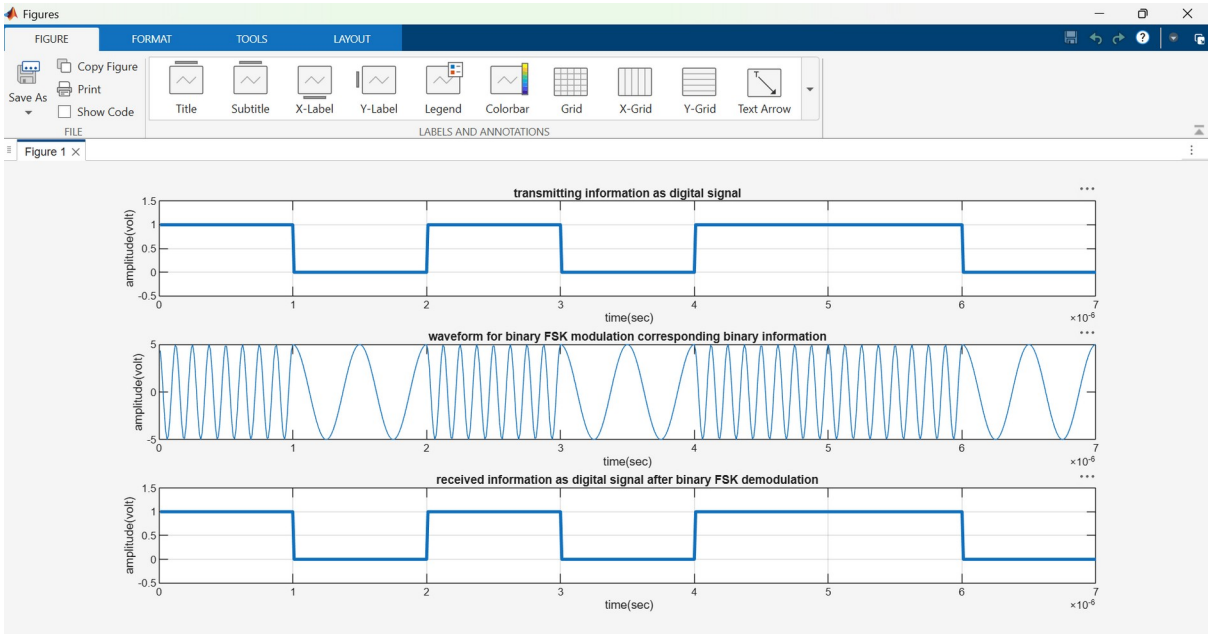
```
    zz2= round(2*z2/bp)
    if(zz1>A/2) % logic level= (0+A)/2 or (A+0)/2 or 2.5 ( in this case)
        a=1;
    else(zz2>A/2)
        a=0;
    end
    mn=[mn a];
end
disp(' Binary information at Receiver :');
disp(mn);

%Representation of binary information as digital signal which archived
%after demodulation
bit=[];
for n=1:length(mn);
    if mn(n)==1;
        se=ones(1,100);
    else mn(n)=0;
        se=zeros(1,100);
    end
    bit=[bit se];
end
t4=bp/100:bp/100:100*length(mn)*(bp/100);
subplot(3,1,3)
plot(t4,bit,'LineWidth',2.5);grid on;
axis([ 0 bp*length(mn) -.5 1.5]);
ylabel('amplitude(volt)');
xlabel(' time(sec)');
title('received information as digital signal after binary FSK demodulation');

```

FIGURE 24: FSK MODULATION AND DEMODULATION CODE PART 2

Practical 5: FSK Modulation and Demodulation Output



4.6 Practical 6: PCM Modulation and Demodulation

```
Enter n value for n-bit PCM system : 24
Enter number of samples in a period : 4
>>
```

FIGURE 25: PCM MODULATION AND DEMODULATION INFORMATION

```
clc;
close all;
clear all;
n=input('Enter n value for n-bit PCM system : ');
n1=input('Enter number of samples in a period : ');
L=2^n;

% Signal Generation
x=0:1/100:4*pi;
y=8*sin(x); % Amplitude Of signal is 8v
subplot(2,2,1);
plot(x,y);grid on;

%Sampling Operation
x=0:2*pi/n1:4*pi; % n1 number of samples have to be selected
s=8*sin(x);

subplot(3,1,1);
plot(s);
title('Analog Signal');
ylabel('Amplitude--->');
xlabel('Time--->');

subplot(3,1,2);
stem(s);grid on;
title('Sampled Signal');
ylabel('Amplitude--->');
xlabel('Time--->');

% Quantization Process
vmax=8;
vmin=-vmax;
del=(vmax-vmin)/L;
part=vmin:del:vmax; %level are between vmin and vmax with difference of del
code=vmin-(del/2):del:vmax+(del/2); % Contain Quantized values
[ind,q]=quantiz(s,part,code); %Quantization process

% ind contain index number and q contain quantized values
l1=length(ind);
l2=length(q);
for i=1:l1
    if(ind(i)~=0)% To make index as binary decimal so started from 0 to N
        ind(i)=ind(i)-1;
    end
    i=i+1;
end
for i=1:l2
    if(q(i)==vmin-(del/2)) % To make quantize value in between the levels
        q(i)=vmin+(del/2);
    end
end
subplot(3,1,3);
stem(q);grid on; %Display the Quantize values
title('Quantized Signal');
ylabel('Amplitude--->');
xlabel('Time--->');

% Encoding Process
figure
code=de2bi(ind,'left-msb'); % Convert the decimal to binary
k=1;
```

FIGURE 26: PCM MODULATION AND DEMODULATION CODE PART 1

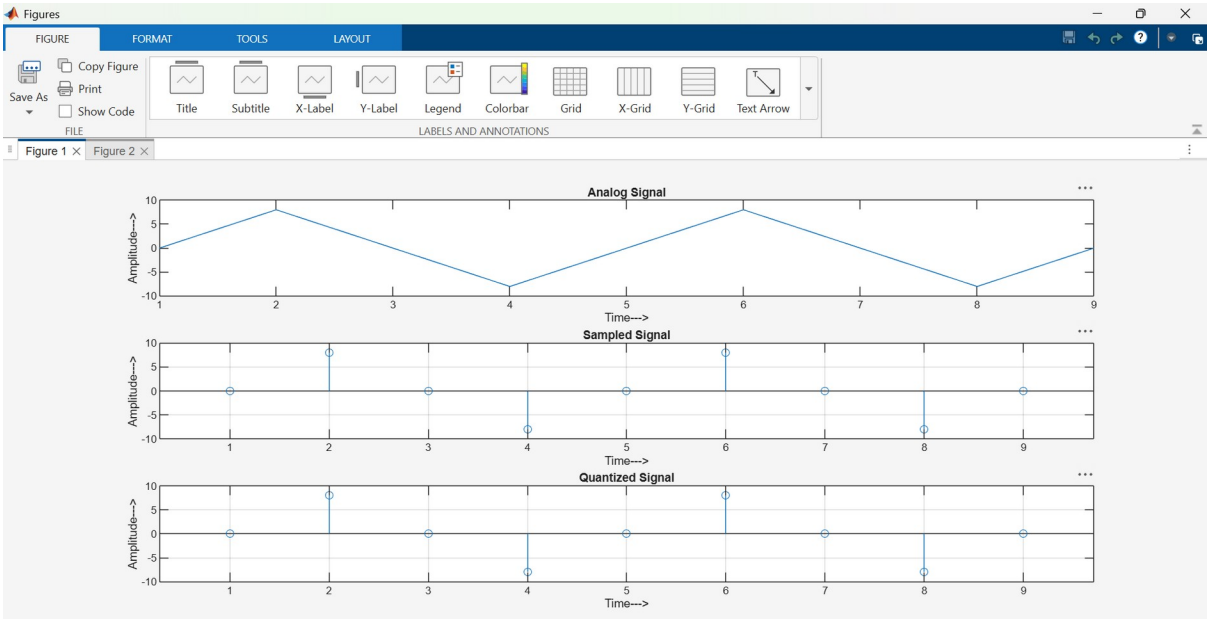
```
for i=1:l1
    for j=1:n
        coded(k)=code(i,j); % convert code matrix to a coded row vector
        j=j+1;
        k=k+1;
    end
    i=i+1;
end
subplot(2,1,1); grid on;
stairs(coded); % Display the encoded signal
axis([0 100 -2 3]); title('Encoded Signal');
ylabel('Amplitude--->');
xlabel('Time--->');

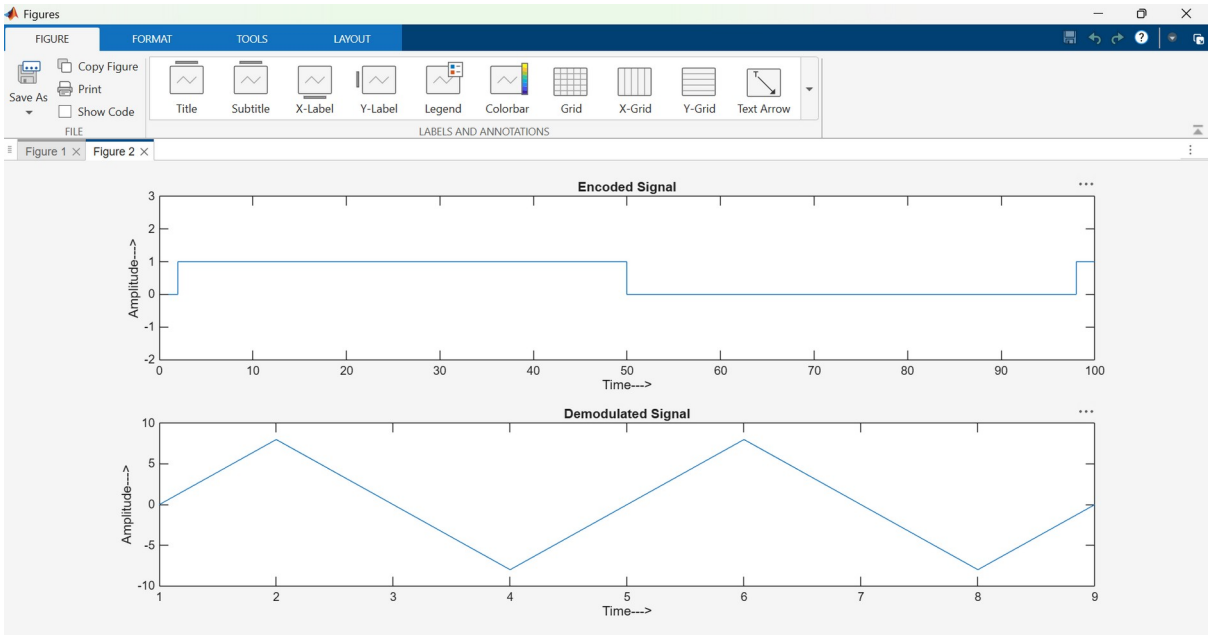
% Demodulation Of PCM signal
qunt=reshape(coded,n,length(coded)/n);
index=bi2de(qunt','left-msb'); %Get back the index in decimal form
q=del*index+vmin+(del/2); %get back Quantized values
subplot(2,1,2); grid on;
plot(q);

% Plot Demodulated signal
title('Demodulated Signal');
ylabel('Amplitude--->');
xlabel('Time--->');
```

FIGURE 27: PCM MODULATION AND DEMODULATION CODE PART 2

Practical 6: PCM Modulation Demodulation Output




Figures
FIGURE
FORMAT
TOOLS
LAYOUT
Save As
Copy Figure
Print
Show Code
FILE
Title
Subtitle
X-Label
Y-Label
Legend
Colorbar
Grid
X-Grid
Y-Grid
Text Arrow
LABELS AND ANNOTATIONS
Figure 1
Figure 2
Encoded Signal
Amplitude---->
Time---->
Demodulated Signal
Amplitude---->
Time---->